

sis early in the design process, it becomes possible to ensure optimal component placement to protect against thermal problems. This, in turn, minimises or eliminates costly rework later. Modern electronic systems incorporate multitudes of components and subassemblies, including circuit boards, fans, vents, baffles, porous plates (such as electromagnetic interference (EMI) shields), filters, cabling, power supplies, disk drives and more. To help designers cope with this complexity, the most advanced thermal modeling solutions provide a comprehensive range of automated software tools and user-friendly menus that provide easier data handling, faster calculations and more accurate results.

Humidity effect

Humidity is the volume or weight of moisture per unit of space—that is, water vapours present in the air. Absolute humidity is a measure of the actual water vapours in the air. It is measured in grams per cubic metre (gm/m^3). Relative humidity is the ratio of absolute humidity to the theoretical maximum for a given temperature and pressure. It is expressed as a percentage. So, if the air holds half of what it could hold, the relative humidity is 50 per cent.

The devastating effects of humidity on electronic equipment are more often underestimated and misunderstood. The consequences of moisture ingress vary with the materials used as follows.

Primary humidity effects. These are the direct result of humidity on equipment or materials and include the following:

Degradation. Humidity may degrade the performance of equipment operating in the infrared band and of some materials such as fabrics, some plastics and cellulose.

De-lamination. De-lamination of composite materials may have a devastating effect on some cheap PCB

base materials.

Dimensional changes. This can result in not only high stress because of bowing/twist but also swelling of fibrous materials.

Fibres. When exposed to humidity, fibres with high moisture-regain values usually not only lose tensile strength and extensibility on the absorption of water but also suffer a fair degree of swelling.

Surface resistivity. On PCBs, reduction in surface resistivity can alter timing circuits (which changes the frequency of oscillator circuits), change the current level in a constant-current source, result in loss of sensitivity, or reduce the input impedance on high-impedance amplifiers.

Metal migration. Metal migration, especially silver migration, occurs between closely spaced conductor lines in the presence of moisture and an applied voltage. It causes reduction of insulation resistance, increase in current leakage and eventually catastrophic shorting. Metal migration is the cause of many microcircuit failures.

Outgassing from PCB laminate. Many PCBs assembled by a number of

different commercial companies have experienced blow holes or voids in solder joints due to outgassing from resins. These defects can be traced to:

1. **Moisture.** One of the main causes of outgassing during the soldering process is release of moisture (absorbed by the resin during long-term storage prior to assembly) in gaseous form due to the extreme temperature during soldering, resulting in poor solder joints, pinholes, blow holes or rupturing of plated-through holes.

2. **Multilayer technology.** The rupturing of a plated-through hole in multilayer boards can also create such problems as disconnection of internal tracks and the resultant high rework costs.

Secondary effects. In the electronics industry,

poor hygiene standards may add sufficient contamination to supply the necessary nutrients to support growth formations. Contamination from the following sources can cause long-term damage that may not be detected during inspection or test:

PCB manufacture. Contamination may take place during actual board production and is usually due to either incomplete curing of the resin or badly applied solder resist.

No-clean fluxes. The input impedance of devices can also get affected by the presence of humidity vapour on any polar contamination in the flux residue. When using hand-soldering techniques, some modern, synthetic no-clean fluxes will not reach deactivating temperature of the flux. If subjected to high humidity for a period of two to three weeks, the result is a white organic salt residue which, although low in ionic contamination, can easily trap moisture. This, in turn, can affect surface resistivity. Hydroscopic nature of these salts may further compound the problem as these present a perfect medium to precipitate dendritic growth when power is applied to the PCB.

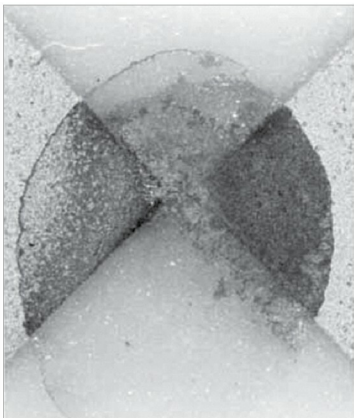


Fig. 2: Metal migration—water-drop test for a silver-platinum, thick-film conductor (complete bridging after 25 minutes, 4V DC)